Statistical Models

UniVR - Dipartimento di Informatica

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1 Introduction

Essentially we can say that this course (SMDS) is based on probability for sequence of random variables. For sequence, we consider a time-ordered set of data.

We can think about temperature forecasting, a set of old values and we want to ease then to forecast. We can write:

- $(X_1, \ldots, X_n)$ an ordered sequence of RV
- $(x_1, \ldots, x_n)$ an ordered sequence of values

We start with discrete RV:

$P_x(x) = P(X = x)$ behind this sequences we have a probabilistic triple

$$(\Omega, \xi, \mathbb{P})$$

We now consider the continuos time case (the most intresting one). A continuos random values (RV) `X` has probability density f_x iff (by definition) there exists a function f_x that:

$$f_x(x) \geq 0$$

$$\int_{\mathbb{R}} f_x(x)\, dx = 1$$

In this case, taking $A \in \xi$, we have

$$\mathbb{P}(X \in A) = \int_A f_x(x)\, dx$$

Not every continuos RV has a density. If we think about a RV `X` taking values in $\mathbb{R}$:

$$\mathbb{P}(X \leq x) = F_x$$

In somesense f_x takes the role of P_x, changing from the discrete (P_x) to the continuos (f_x)